Name: _____ Class: _____

JURONG PIONEER JUNIOR COLLEGE
JC2 Theory Practice Set 2 (2026)

COMPUTING **9569/01**

Higher 2

Paper 1 (Written) 3 hours

Additional materials: Answer Booklet

READ THESE INSTRUCTIONS FIRST

Answer **all** the questions.

Approved calculators are allowed.

You are reminded of the need for clear presentation in your answers.

Answer papers will be provided with the question paper.

Write your name and civics class on all the work that you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Write your answers to every question on a fresh page of paper.

The number of marks is given in brackets [ ] at the end of each question or part question.

The total number of marks for this paper is 100.

This document consists of 9 printed pages.

1 CrowdTaskSG is a government platform on which members of the public take part in paid surveys, quests and A/B testing that help government agencies improve their services. It is open to Singapore citizens and permanent residents aged 15 and above who have a Singpass account.

In a proposed design, each participant is given a 7-character participant ID made up of the letter C, five digits and a check letter, for example C40213J. The check letter is calculated as follows:

1. Multiply the five digits by the weights 6, 5, 4, 3 and 2 respectively.
2. Add the five products to give `Total`.
3. Calculate `R ← Total MOD 11`.
4. The check letter is the character at position `R` of the string "ABCDEFGHJKL", where the first character is at position 0.

For example, for C40213: 4×6 + 0×5 + 2×4 + 1×3 + 3×2 = 41, 41 MOD 11 = 8, so the check letter is "J".

(a) Explain the difference between data validation and data verification. Give an example of each that could be used when a participant registers on CrowdTaskSG. [2]

(b) Calculate the check letter for the participant ID that begins C57381. Show your working. [3]

(c) A participant with the ID C57381 followed by the correct check letter types C75381 followed by the same check letter. Explain how the check letter allows this error to be detected. [2]

(d) Write pseudo-code for the function `IsValidID(ID : STRING) RETURNS BOOLEAN` that returns TRUE only if:

- `ID` is exactly 7 characters long
- the first character is "C"
- the characters at positions 1 to 5 are all digits
- the last character is the correct check letter.

Positions start at 0. You may use `ID[i]` to return the character at position `i`, and `ASC(ch)`, which returns the ASCII code of `ch`. The ASCII codes for "0" to "9" are 48 to 57. [5]

(e) State a suitable value of age to use as extreme test data when testing the age requirement for CrowdTaskSG, and explain your choice. [1]

2 OneMap, the national map platform of the Singapore Land Authority (SLA), has a Find Nearby Amenities feature. For a chosen location, the distances in metres to eight nearby clinics are stored in an array:

```
640   210   980   150   420   870   330   510
```

(a) A quicksort that always uses the first element of each sub-list as the pivot is used to sort the distances into ascending order. Use annotated diagrams to show how the quicksort sorts the distances. [3]

(b) An insertion sort is used instead. State the contents of the array after each of the first three passes of the insertion sort. [3]

(c) When a user pans the map slightly, one new distance is added to the end of a list of 2000 distances that is already in ascending order, and the list must be sorted again. Explain why an insertion sort is more suitable than this quicksort for this task. [2]

(d) OneMap's address search stores 120 000 six-digit postal codes in an array in ascending order.

(i) State the maximum number of postal codes that need to be compared when a binary search is used on this array. [1]

(ii) Using Big-O notation, compare the worst-case time complexity of a linear search and a binary search on this array. [2]

(e) The postal codes could instead be stored in a hash table. Explain why searching a well-designed hash table is usually faster than a binary search, and state one disadvantage of the hash table compared with the sorted array. [2]

3 PlayTime was developed by Open Government Products (OGP) during its Hack for Public Good hackathon. It helps parents discover activities for their children. Parents enter the child's age, a location or postal code, and the child's interests. Listings come from both government and private organisers.

A prototype stores all the listings in a single table, `ActivityListing`. Part of the table is shown below.

Activity ID	Title	Activity Date	Min Age	Max Age	Organiser ID	Organiser Name	Organiser Type	Postal Code	Venue Name	Interests
A01	Junior Nature Walk	2026-11-07	5	9	G02	NParks	Government	259569	Singapore Botanic Gardens	Nature, Outdoor
A02	Robotics Taster	2026-11-08	7	12	P15	BrickLab	Private	469662	Heartbeat @ Bedok	Science
A03	Storytime and Craft	2026-11-08	3	6	G07	NLB	Government	469662	Heartbeat @ Bedok	Reading, Art
A04	Family Kite Day	2026-11-14	4	12	P22	SkyHigh Kites	Private	018951	Marina Barrage	Outdoor

(a) Explain why `ActivityListing` is not in first normal form (1NF). Refer to the data in your answer. [2]

(b) A table description can be expressed as:

TableName(Attribute1, Attribute2, Attribute3, ...)

The primary key is indicated by underlining one or more attributes. Foreign keys are indicated by using a dashed underline.

The data is to be stored in five tables in third normal form (3NF): `Organiser`, `Venue`, `Activity`, `Interest` and `ActivityInterest`. Each interest is to be given an interest code. Write table descriptions for the five tables. [5]

(c) Draw an entity-relationship (ER) diagram to show the five tables and the relationships between them. [3]

(d) Write an SQL query to display the title, date and venue name of every activity that is suitable for a child aged 6 and has the interest 'Outdoor'. The activities should be displayed with the earliest date first. [4]

4 GovTech encourages Singpass users to create a passkey. Information shown to the public includes the following:

- Passkeys are a safer way to log in and offer stronger protection against phishing scams.
- Passkeys only work on the legitimate Singpass login domain and cannot be used on fake websites.
- Users authenticate using their biometrics or 6-digit Singpass app passcode during passkey logins.
- Passkeys are stored securely on the device.
- No passwords or OTPs are entered during passkey logins, so there is nothing for scammers to steal.

A passkey uses asymmetric (public key) cryptography. When a user creates a passkey, the user's phone generates a key pair. The private key is kept on the phone and the public key is sent to the Singpass server, which stores it with the user's account.

(a) Explain why it is safe for the Singpass server to store the public key, but the private key must never leave the phone. [2]

(b) At each login, the Singpass server sends a new random value, called a challenge, to the phone. Describe how the challenge and the stored public key are used to authenticate the user. [3]

(c) Explain why a passkey gives better protection against a phishing website than logging in with a password and a one-time password (OTP). [3]

(d) Explain why a passkey login, in which the user unlocks the passkey with a fingerprint or the Singpass app passcode, is a form of multi-factor authentication. [2]

5 OneMap offers Barrier-Free Access Routing for persons with disabilities, wheelchair users, seniors and families with prams. A student models a small area as a two-dimensional array `Map` of BOOLEAN values. `Map[r][c]` is TRUE if the square in row `r` and column `c` can be used by a wheelchair, and FALSE if it cannot (for example, it has steps).

A wheelchair user starts at `Map[0][0]` and can only move one square South (to row `r + 1`) or one square East (to column `c + 1`) at a time. The recursive function `Routes` returns the number of different routes from `Map[0][0]` to `Map[r][c]`.

```
01  FUNCTION Routes(Map, r : INTEGER, c : INTEGER) RETURNS INTEGER
02     DECLARE Total : INTEGER
03     IF Map[r][c] = FALSE THEN
04        RETURN 0
05     ENDIF
06     IF r = 0 AND c = 0 THEN
07        RETURN 1
08     ENDIF
09     Total ← 0
10     IF r > 0 THEN
11        Total ← Total + Routes(Map, r - 1, c)
12     ENDIF
13     IF c > 0 THEN
14        Total ← Total + Routes(Map, r, c - 1)
15     ENDIF
16     RETURN Total
17  ENDFUNCTION
```

(a) `MapA` has 2 rows and 3 columns, and every element is TRUE. Draw a trace tree for the call `Routes(MapA, 1, 2)`. Show the parameters `r` and `c` of every call and the value returned by every call. [3]

`MapB` has 3 rows and 4 columns. T represents TRUE and F represents FALSE.

	Column 0	**Column 1**	**Column 2**	**Column 3**
Row 0	T	T	T	F
Row 1	T	F	T	T
Row 2	T	T	T	T

(b) State the value returned by `Routes(MapB, 2, 3)`. [1]

(c) Explain the purpose of the two base cases in lines 03 to 08. [2]

(d) Explain why the function `Routes` becomes very slow when it is used on a large map. [2]

(e) Write pseudo-code for an iterative function `RoutesIter(Map, R, C)` that returns the same value as `Routes(Map, R, C)`. Your function should use a two-dimensional array `Count` to store the number of routes to each square. [4]

6 Spot the Scam! is a browser-based game, developed by the Centre for Advanced Technologies in Online Safety (CATOS) with the Singapore Institute of Technology (SIT), in which players decide whether messages are scams. The game is implemented using object-oriented programming.

Every message has a message ID, a sender and its content. Every message has a method `RedFlagCount()` that returns the number of red flags in the message. For every type of message, there is one red flag if the content contains the word "urgent".

There are three types of message:

- SMS: also stores the sender's phone number. There is one extra red flag if the number starts with "+" but not with "+65".
- Email: also stores the sender's domain, the domain of the link in the email, and whether the email has an attachment. There is one extra red flag if the link domain is different from the sender's domain, and one extra red flag if there is an attachment.
- Call: a simulated voice call. It also stores the organisation the caller claims to be from, and whether the caller asks for a one-time password (OTP). There are two extra red flags if the caller asks for an OTP.

(a) Draw a class diagram for the game. Show the superclass, the subclasses, the inheritance relationships, the attributes and appropriate methods, including any methods that are overridden. Show clearly which attributes and methods are private and which are public. [5]

(b) Write pseudo-code for the class `Email`. Your class must include:

- a constructor that makes use of the constructor of the superclass
- the method `RedFlagCount()`, which must make use of the superclass's `RedFlagCount()` method.

[4]

(c) A new type of message, `DeepfakeVideo`, is to be added to the game. Explain how polymorphism allows this to be done without changing the code that calculates a player's score from the red flags of every message in a round. [2]

7 At the Digital for Life Festival, visitors to the National Library Board (NLB) booth solve a mystery game and submit their answers on their mobile phones using an online form created with FormSG, a government form-builder web application. The phones connect to the internet through the festival's Wi-Fi network.

(a) With reference to the submission of a visitor's answers, explain the roles of the client and the server in a client-server architecture. [2]

(b) A visitor opens the link to the form, which begins with https://form.gov.sg. Describe how the visitor's phone finds the IP address of the FormSG server. [2]

(c) Give two reasons why the booth uses a web application for the form instead of asking visitors to install a native application. [2]

(d) Explain why communication protocols are needed for the visitor's phone to communicate with the FormSG server. [1]

(e) During the festival, a server that hosts an online registration page receives a very large number of requests from thousands of computers that have been infected with malware.

Explain how this attack can make the registration page unavailable, why a firewall that filters traffic may not be able to stop it, and how an intrusion prevention system (IPS) could help. [3]

(f) At the Cyber Security Agency (CSA) booth, visitors learn about malware. Explain the difference between a virus and a worm. [2]

8 In October 2026, news media reported that a small developer had used a generative AI model, directing eight AI coding agents working in parallel, to produce a suite of free, open-source applications that closely imitate the look and main features of Adobe's Creative Cloud applications, such as Photoshop and Lightroom. The developer of a similar Photoshop-style editor said that it cost about US$2,000 in AI usage to build. Commentators asked whether this was a "SaaSpocalypse 2.0", in which AI makes subscription software (Software as a Service) cheap to copy.

The applications are downloaded and installed from a code-sharing website. One of the applications was reported to have more than 1,700 automated tests, and 307 of 309 test image files were opened and saved again without any change.

(a) Explain one risk to a user who downloads such an application from an unofficial source, and explain how a digital signature on the download could reduce this risk. [3]

(b) Eight AI agents made changes to the same program code at the same time. Explain why version control and naming conventions are needed in this situation. [3]

(c) (i) Explain why passing all of the automated tests does not prove that the application is free of errors. [1]

(ii) Give one example of normal test data and one example of abnormal test data that could be used to test the feature that opens image files, and state the expected outcome for each. [2]

(d) Discuss the social, ethical, legal and economic issues that arise when generative AI can be used to produce working copies of commercial software cheaply. [6]

END OF PAPER